

MILESTONE 1

System Design & Architecture

Deliverables:

- Full system architecture (MAIN + REMOTE separation clearly defined)
- Power flow design (240V → DC → battery → load)
- RF communication architecture + pairing logic
- Sensor and override logic design (state machine concept)
- Component selection with justification
- Initial schematic draft (block-level + partial circuit structure)

Expected Outcome:

- How both PCBs interact
- How power is managed safely
- How control states transition (Auto / Manual / Sensor override)
- All key components selected and justified before PCB design begins

2. Schematic Design

Deliverables:

- **Complete electrical schematics for:**
 - Main 240V AC Night Light PCB
 - RF Wireless Remote PCB
- **Full circuit design for all subsystems:**
 - AC input protection (fuse, isolation stage)
 - AC–DC conversion system
 - Battery charging and backup switching circuit
 - PIR motion + ambient light sensing circuit
 - RF receiver and decoding logic (main board)
 - RF transmitter circuit (remote board)

Expected Outcome:

- Correct separation of high-voltage and low-voltage domains
- Verified interaction between sensors, RF system, and power control
- Complete circuit logic before PCB layout begins

PRICE: \$200

MILESTONE 2

Stage 3: PCB Layout Design

Deliverables:

- Complete PCB layout for both boards
- **Main PCB layout:**
 - Circular board design optimized for enclosure constraints
 - Proper high-voltage creepage and clearance spacing
 - Strategic placement of AC section vs low-voltage section
 - Thermal-aware LED placement for uniform light output
- **RF Remote PCB layout:**
 - Compact handheld form factor design
 - Optimized button placement for usability
 - RF antenna positioning for maximum transmission reliability
- **Component placement optimization:**
 - Power section isolation from sensitive control circuitry
 - Noise reduction layout practices for RF and sensor stability
- Design rule check (DRC) and manufacturability validation

Expected Outcome:

- Safe physical implementation of 240V AC system
- Reliable RF communication performance
- Compact and manufacturable circular form factor
- Clean separation between power, signal, and RF domains
- Ready for Gerber export and fabrication

Stage 4: Manufacturing Package and Two Rounds of Revision

Deliverables:

- Gerber files + drill files
- Full BOM with sourcing links (DigiKey / LCSC / Mouser)
- Datasheets for all critical components
- Assembly-ready documentation

Expected Outcome:

- PCB fabrication

- SMT assembly
- Prototype testing

PRICE \$200

TOTAL PRICE \$400